MES COLLEGE OF COMMERCE, ARTS AND SCIENCE

(Affiliated to Dr. Manmohan Singh Bengaluru City University)

NAAC Accredited with Grade 'A'

15th Cross, Vidyasagara Prof. MPL Sastry Road, Malleswaram, Bengaluru- 560003

A SYNOPSIS

ON

INVORA – AI Powered Personal Study Assistant

Submitted in partial fulfillment of the requirements for the award of the Degree of

BACHELOR OF COMPUTER APPLICATIONS

in

DEPARTMENT OF COMPUTER APPLICATIONS

of

Dr. Manmohan Singh Bengaluru City University

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SUBMITTED BY

Chiranth G

U18EM24S0076

Gagan Gowda C

U18EM24S0014

Bharath S

U18EM24S0094

Under Supervision of

Guide Name : **Mrs. Chethana E**

Assistant Professor

Department of Computer Applications

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DR. MANMOHAN SINGH BENGALURU CITY UNIVERSITY

(Duration: _ 2026 to _ 2027)

DEPARTMENT OF COMPUTER APPLICATIONS

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BATCH 2026 – 2027

Sl No.	Contents	Page No.
1.	Introduction	0-1
2.	Objective	0-1
3.	Problem Statement	0-1
4.	Hardware and Software Requirements	1-2
5.	Existing System	1-2
6.	Proposed System	2-3
7.	Methodology	2-3
8.	Conclusion	3-4

1. Introduction

Artificial Intelligence (AI) is transforming education by making learning more interactive and efficient. This project, INVORA – AI-Powered Personal Study Assistant, helps students upload PDF study materials and access AI-powered learning features. It generates notes, summaries, quizzes, and answers questions based on the uploaded content. The project aims to simplify exam preparation and improve students' overall learning experience.

2. Objective

- To design and develop an AI-powered personal study assistant.
- To simplify the process of studying from PDF documents.
- To generate AI-based notes, summaries, and quizzes from uploaded study materials.
- To provide a secure and user-friendly learning platform.
- To improve students' learning efficiency and exam preparation.
- To track student's learning progress through an interactive dashboard.

3. Problem Statement

Students often rely on traditional methods of studying, such as reading lengthy PDF documents and preparing notes manually, which can be time-consuming and less effective. Existing learning systems provide limited support for interactive learning, quick content understanding, and self-assessment. As a result, students face difficulties in efficient exam preparation and tracking their learning progress. Therefore, there is a need for an AI-powered study assistant that enables intelligent PDF interaction, automatic note generation, quiz creation, and progress monitoring to improve the overall learning experience.

4. Hardware and Software Requirements

This section specifies the resources required to develop and execute the project.

4.1 Hardware Requirements

Processor	Intel Core i3 or above
RAM	8 GB or above
Storage	256 GB SSD/HDD or above
Internet Connection	Required
Display	1366 × 768 resolution or higher

4.2 Software Requirements

Operating System	Windows 10/11
Frontend	HTML, CSS, JavaScript
Backend	Python (Flask)
Database	MySQL
AI Integration	Google Gemini API
IDE/Code Editor	Visual Studio Code
Web Browser	Google Chrome / Microsoft Edge

5. Existing System

The existing system mainly depends on manual study methods, where students read PDF materials and prepare notes themselves. It lacks intelligent features for quick understanding and self-assessment. As a result, learning becomes time-consuming and less efficient.

➢ Current Working Procedure

Students study by reading PDF materials, preparing notes manually, and creating practice questions on their own.

➢ Problems Faced

- Time-consuming study process.
- Manual note preparation.
- Limited self-assessment.

➢ Limitations

- No AI assistance.
- No automatic notes or quizzes.
- No progress tracking.

➢ Need for Improvement

An AI-powered system is needed to provide smart learning, automatic notes, quizzes, and progress tracking for effective study.

6. Proposed System

The proposed system uses Artificial Intelligence to make learning more efficient and interactive. It allows users to upload PDFs, generate notes and quizzes, and ask questions based on the uploaded content. The system also tracks learning progress to support

➢ Features

- User Login & Registration
- PDF Upload & Management
- AI Chat, Notes, and Quiz Generation
- Progress Tracking Dashboard

➢ Advantages

- Reduces manual study effort.
- Provides AI-based learning support.

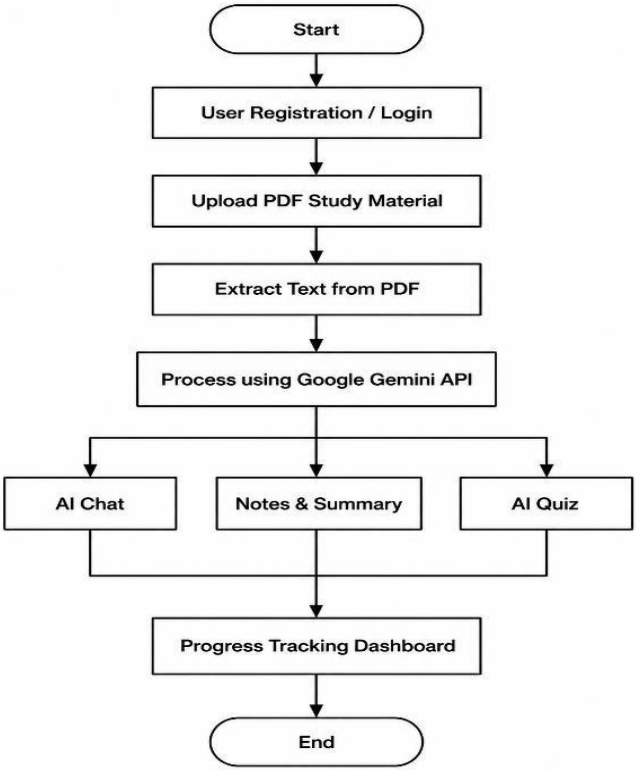
➢ Benefits

- Saves time.
- Improves learning and exam preparation.

➢ Future Enhancement

- Mobile application support.
- Voice and multi-language support.

7. Methodology



8. Conclusion

The proposed project, INVORA – AI-Powered Personal Study Assistant, successfully achieves its objective of providing a smart and interactive learning platform for students. The system enables users to upload PDF study materials, interact with AI, generate notes, summaries, quizzes, and track their learning progress efficiently. It reduces manual effort, saves time, and enhances the overall learning experience. In the future, the system can be further enhanced by adding features such as voice-based assistance, multilingual support, mobile application integration, and cloud storage for improved accessibility and usability.